RODENT OPERANT CHAMBER BUCKET

MINOR PROJECT REPORT

Submitted in partial fulfillment of the requirement for the

award of degree of

BACHELOR OF ENGINEERING

IN

ELECTRICAL AND ELECTRONICS ENGINEERING

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UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY

PANJAB UNIVERSITY

CHANDIGARH

Candidates' Declaration

We hereby declare that the entire project work embodied in this minor project entitled ***"RODENT OPERANT CHAMBER BUCKET"***, towards the partial fulfillment of the requirement for the degree of **B.E.** in **Electrical and Electronics Engineering**, is an authentic bonafide record of our work carried out under the supervision of ***Dr. Yajvender Pal Verma***, Assistant Professor, Department of Electrical and Electronics Engineering, UIET, Panjab University, Chandigarh.

The work presented in this project has not been submitted by us for the award of any other degree of this or any other University. We have taken care in all respects to honour the intellectual property rights and we have acknowledged the contribution(s) of others for using them for this academic purpose.

Date: 28th November 2019

Place: Chandigarh

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Certificate

This is to certify that **Mr. Abhinav Singh, Mr. Arnav Sharma** and **Mr. Manik Dalal** completed their minor project under my supervision and the given declaration made by the students is correct and true to the best of my knowledge.

_____

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Acknowledgement

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LIST OF ABBREVIATIONS

- RO Bucket — Rodent Operant Bucket
- IDE — Integrated development environment
- IR Sensors — Infrared Object Detection Sensors
- In — Inches
- OB — Operational Behaviour
- SV — Solenoid Valve

ABSTRACT

"The operant conditioning chamber is a cornerstone of animal behavioural research. Operant boxes are used to assess learning and motivational behaviour in animals, particularly for food and drug reinforcers. However, commercial operant chambers cost several thousands of dollars. We have constructed the Rodent Operant Bucket (ROBucket), an inexpensive and easily assembled open-source operant chamber based on the Arduino microcontroller platform, which can be used to train mice to respond for sucrose solution or other liquid reinforcers. The apparatus contains two nose pokes, a drinking well, and a solenoid-controlled liquid delivery system. RO Bucket can run fixed ratio and progressive ratio training schedules, and can be programmed to run more complicated behavioural paradigms. Additional features such as motion sensing and video tracking can be added to the operant chamber through the array of widely available Arduino-compatible sensors. The design files and programming code are open source and available online for others to use."

RODENT OPERANT CHAMBER BUCKET

Chapter 1. Introduction

1.1. Animal Behaviour Scenario

Animal behavior describes the mental capacities of non-human animals and the study of those capacities. The field developed from comparative psychology, including the study of animal conditioning and learning. It has also been strongly influenced by research in ethology, behavioral ecology, and evolutionary psychology, and hence the alternative name cognitive ethology is sometimes used. Many behaviors associated with the term *animal intelligence* are also subsumed within animal cognition.

Researchers have examined animal cognition in mammals, birds, reptiles, fish and invertebrates.

Several experiments cannot be readily reconciled with the belief that some animal species are intelligent, insightful, or possess a theory of mind.

Jean-Henri Fabre (1823-1915), setting the stage for all subsequent experiments of this kind, argued that insects “obey their compelling instinct, without realizing what they do.” For instance, to understand that she can grab her paralyzed prey by a leg instead of an antenna is utterly beyond the powers of a sand wasp. ““Her actions are like a series of echoes each awakening the next in a settled order, which allows none to sound until the previous one has sounded.” Fabre’s numerous experiments led him, in turn, to the view that scientists often try to “exalt animals” instead of objectively studying them.

C. Lloyd Morgan's (1852-1936) observations suggested to him that prima facie intelligent behavior in animals is often the result of either instincts or trial and error. For instance, most visitors watching Morgan’s dog smoothly lifting a latch with the back of its head (and thereby opening a garden gate and escaping) were convinced that the dog’s actions involved thinking. Morgan, however, carefully observed the dog’s prior, random, purposeless actions and

argued that they involved “continued trial and failure, until a happy effect is reached,” rather than “methodical planning.”

E. L. Thorndike (1874 –1949) placed hungry cats and dogs in enclosures “from which they could escape by some simple act, such as pulling at a loop of cord.” Their behavior suggested to him that they did not “possess the power of rationality.” Most books about animal behavior, Thorndike wrote, “Do not give us a psychology, but rather a eulogy of animals.”

Although Wolfgang Köhler's experiments are often cited as providing support for the animal cognition hypothesis, his book is replete with counterexamples. For instance, he placed chimpanzees in a situation where they could only get bananas by removing a box. The chimpanzee, Köhler observed, “has special difficulty in solving such problems; he often draws into a situation the strangest and most distant tools, and adopts the most peculiar methods, rather than remove a simple obstacle which could be displaced with perfect ease.”

Daniel J Povinelli and Timothy Eddy of the University of Louisiana showed that chimpanzees, when given a choice between two food providers, were just as likely to beg food from a person who could see the begging gesture as from a person who could not, thereby raising the possibility that chimpanzees do not understand that people see.

Moty Nissani of Wayne State University trained Burmese logging elephants to lift a lid in order to retrieve food from a bucket. The lid was then placed on the ground alongside the bucket (where it no longer obstructed access to the food) while the treat was simultaneously placed inside the bucket. All elephants continued to toss the lid before retrieving the reward, thus suggesting that elephants do not grasp simple causal relationships.

1.2. Motivation for this Work

The study of animal behavior begins with understanding how an animal's physiology and anatomy are integrated with its behavior. Both external and internal stimuli prompt behaviors — external information (e.g., threats from other animals, sounds, smells) or weather and internal information (e.g., hunger, fear). Understanding how genes and the environment come together to shape animal behavior is also an important underpinning of the field. Genes capture the evolutionary responses of prior populations to selection on behavior. Environmental flexibility gives animals the opportunity to adjust to changes during their own lifetime.

Scientists are drawn to the study of animal behavior for varied reasons and the field is extremely broad, ranging from research on feeding behavior and habitat selection to mating behavior and social organizations. Many scientists study animal behavior because it sheds light on human beings. Research on non-human primates, for instance, continues to offer valuable perspectives into the causes and evolution of individual, social, and reproductive human actions. Understanding why some animals help others at the potential cost of their own survival and reproduction, for example, not only gives us insight into their behavior but could also potentially help us to understand the underpinnings of our species' ideas of altruism and sacrifice.

Operant training can be used to test learning and motivation in animals, and has been used extensively in studies of food reinforcement, drug addiction, and learning and memory. However, commercially available operant chambers cost ~ INR 1,40,000–3,50,000 per chamber, and thus may prevent researchers and educators from performing operant behavioural experiments. Even in well-funded research environments, this cost can make high-throughput experiments involving dozens of operant chambers impractical.

To address the issue of cost, we have constructed an inexpensive and easily assembled operant chamber based on the Arduino electronics platform. Arduino input/output (I/O) boards are ideal laboratory tools because they can receive and transmit transistor-transistor-logic (TTL) signals on a millisecond timescale without a computer interface (D'Ausilio, 2012). Arduino programs – called sketches – are written in a simple language with the Arduino integrated development environment (IDE) software. The bootloader on the Arduino board allows users to upload sketches

from a computer via a universal serial bus (USB). In addition, there are many publicly available, open-source libraries, which are sets of programs that define common functions. Libraries allow users to more easily execute complex commands thus expanding the capabilities of the Arduino programming language and hardware.

1.3. Outline of this Report

This report contains detailed information for construction of a low cost RO Bucket for animal behaviour analysis.

The report is divided into 5 Chapters which look into the various aspects of the knowledge required and construction methodology.

I. **Introduction** contains brief information about the current animal behaviour scenario and why it is needed to study animal behaviour. It also contains what factors led to the development of this project.

II. **Literature Survey** shows the analyses and researches made in the field of your interest and result already published, taking into account the various parameters of the project and the extent of the project. It contains information about animal behaviour, previous developments of RO Bucket, microcontrollers. We also devised the problem statement and objectives of the work to be completed along with the jest of the whole survey.

III. **System Description** contains the information about the instruments required. It also have the information about the hardware, control mechanism and data logger required for the objectives of the project.

IV. **Performance of the System in Real Time Environment** contains the details about the experimentation on live rodents and analysis of its behaviour during a specified time along with recorded data which is analysed graphically.

V. **Conclusions and Future Scope** contains information about the jest of the report and future modifications to be carried out during prototype development phase.

Chapter 2. Literature Survey

2.1 Introduction

The RO Bucket chamber consists of a four-quart square plastic container (6.5 in × 6.5 in × 8 in.) and a simple input/output circuit. The chamber is equipped with three photo interrupter sensors, which detect interactions with the nose pokes and drinking well. The nose pokes flank the drinking well and are indistinguishable from each other except that responding on the designated active poke results in sucrose delivery whereas responding on the inactive poke produces no consequence. Responses on both pokes and the drinking well are logged to the SD card. The drinking well is distinctive in that a blunt needle descends into the opening, which is enclosed from the bottom with a clear plastic square.

The infrared sensors send a constitutively high (5V) TTL signal to the Arduino. A nose-poke response involves a mouse breaking the infrared photobeam of a photo interrupter board with its nose. When a mouse responds on a nose poke, a blow signal is sent to the Arduino, triggering the sucrose delivery process. To avoid false responses, all reinforcement schedules require the mouse to remove his nose between pokes, and the FR1 program includes a 1-s timeout in which the nose poke is inactive following each active signal.

After receiving a low signal from the active nose poke, Arduino sends a 5V output signal to trigger a solenoid valve through a relay module. The signal causes the solenoid to open for 75 ms, allowing approximately 60 μL of sucrose solution to be dispensed into the drinking well. As each behavioural event occurs, the Arduino timestamps the event to a file on the SD card and updates the trial information displayed on the LCD keypad shield.

2.2 Overview of the Work

The work of RO bucket can be summarized as follows:

I. **Hardware** is the most important part of the project as it is important to make an ergonomic, animal safe along with equipment required for the objectives to be achieved .Positioning of equipments and cost effectiveness is also an important aspect.

II. **Control Mechanism** is the heart of the project which will analyse all the data and send appropriate signals to take the required control actions to take place. It also helps to display the required data for analysis.

III. **Data Logger** is also required as analysis require systematic retrieval of data along with a presentable data format. It provides real time data in a very appropriate and executable form to maintain graphs and tables.

IV. **Performance of the System in Real Time Environment** is the most important objective as it helps to detect the faults and anomalies which needed to be rectified in order to get genuine and accurate data.

2.3 Animal Behaviour

Behavior usually implies to animal's reactions. It defines 'what' an animal does and 'how' it does it. It results as a reaction to stimulus. Stimulus is a detectable change in the animal's internal or external environment.

- Hunger
- Sound
- Pain
- Visual cues

Example:

MORRIS WATER MAZE MODEL FOR SPATIAL LEARNING AND MEMORY

- Richard Morris developed the Morris water maze
- It consists a tank of water containing a platform just below surface of water. The water is made opaque by adding white powder.
- A rat was placed in the tank. Due to their natural tendency to avoid water, the animal tries to locate the way out.
- It swims across the tank until it stumbles upon the platform.
- On subsequent trials, the rat finds more direct and quicker pathway to reach the platform.

Fig 1. Animal Behaviour Models

2.4 Rodent Operant Bucket

- Mice are trained in RO Bucket with active and inactive poke to meet the acquisition criteria for addiction behavior.

OPERATION	RESPONSE
Nose Poking in Active	**Delivery of Sucrose**
Nose Poking in Inactive	**No Response**

Table 1. RO Bucket Sensor Response

- In this way, a researcher may reinforce nose-poking behavior in mice by delivering sucrose solution each time the mouse activates a trigger by poking with its nose.
- If a behavior occurs with a higher frequency following an outcome (such as delivery of sucrose) the behavior has been reinforced whereas if it occurs with lower frequency it has been punished.
- Operant training can be used to test learning and motivation in animals, and has been used extensively in studies of food reinforcement, drug addiction, and learning and memory.

Above Specified experiment is performed and data is recorded in RO Bucket.

2.5 Microcontrollers

As of 2008, there are several dozen microcontroller architectures and vendors including:

- ARM core processors (many vendors)
- Microchip Technology Atmel AVR (8-bit), AVR32 (32-bit), and AT91SAM (32-bit)
- Cypress Semiconductor's M8C core used in their PSoC (Programmable System-on-Chip)
- Freescale ColdFire (32-bit) and S08 (8-bit)
- Freescale 68HC11 (8-bit), and others based on the Motorola 6800 family
- Intel 8051, also manufactured by NXP Semiconductors, Infineon and many others

Many others exist, some of which are used in very narrow range of applications or are more like applications processors than microcontrollers. The microcontroller market is extremely fragmented, with numerous vendors, technologies, and markets. Note that many vendors sell or have sold multiple architectures.

In this device, low computational ability and low level of transmission speed is required. Meagre Data Memory is required for data logger. An open source and robust architecture is required for the controller as well as the cost of the controller should be less. Considering the above mentioned requirements **ARDUINO UNO** is best suited option for the device.

2.6 Problem Statement

"*To develop a cost-efficient device to overlook operational behaviour of rodents and analyse the data in tabular format.*"

2.7 Objectives of the Work

The following objectives are derived after carefully studying the literature required for the RO Bucket:

1. Detect the movement of the rodent in the nose pokes of the device
2. Dispense a suitable reward at the reward nose poke of 60µL drop size.
3. Behaviour Data is to be stored and displayed in proper graphical format
4. The cost of the device should be reduced as compared to other competitors.
5. The design should be robust and non-corrosive.

2.8 Summary

After studying and analysing the required literature to conceptualize the RO Bucket ,we have defined a particular test condition.(For experiment test conditions refer Table 1)We have selected arduino as our main computing unit and IR sensors the detecting device for the movement of the rodents as rodent don't find IR an interference to their naked eye. The size and performance parameters are also defined and a volume of 60ML alcohol / glucose is set as a perfect reward. SD card is chosen as perfect external storage device. Closed loop control system architecture is used in the system. ThingSpeak is the required software to show recorded data in graphical format. Solenoid valve controlled with a 5V relay is determined to be a suitable fluid dispense mechanism with a manual droplet flow limiter.

The above set parameters are tend to change during the performance analysis period to enhance the capability of the device.

Chapter 3. System Description

3.1 Introduction

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The photo interrupter sensors send a constitutively high (5V) TTL signal to the Arduino. A nose-poke response involves a mouse breaking the infrared photo beam of a infrared sensors with its nose. When a mouse responds on a nose poke, a low signal is sent to the Arduino, triggering the sucrose delivery process. To avoid false responses, all reinforcement schedules require the mouse to remove his nose between pokes, and the FR1 program includes a 1-s timeout in which the nose poke is inactive following each active signal.

After receiving a low signal from the active nose poke, Arduino sends a 5V output signal to trigger a solenoid valve through a relay module. The signal causes the solenoid to open for 75 ms, allowing approximately 60 μL of sucrose solution to be dispensed into the drinking well. As each behavioral event occurs, the Arduino timestamps the event to a file on the SD card and updates the trial information displayed on the LCD keypad shield.

3.1.1. Arduino

Arduino is an open-source hardware and software company, project and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices. Its products are licensed under the GNU Lesser General Public License (LGPL) or the GNU General Public License (GPL), permitting the manufacture of Arduino boards and

software distribution by anyone. Arduino boards are available commercially in preassembled form or as do-it-yourself (DIY) kits.

Arduino board designs use a variety of microprocessors and controllers. The boards are equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards ('shields') or breadboards (For prototyping) and other circuits. The boards feature serial communications interfaces, including Universal Serial Bus (USB) on some models, which are also used for loading programs from personal computers. The microcontrollers can be programmed using C and C++ programming languages. In addition to using traditional compiler tool chains, the Arduino project provides an integrated development environment (IDE) based on the Processing language project.

The Arduino project started in 2005 as a program for students at the Interaction Design Institute Ivrea in Ivrea, Italy, aiming to provide a low-cost and easy way for novices and professionals to create devices that interact with their environment using sensors and actuators. Common examples of such devices intended for beginner hobbyists include simple robots, thermostats and motion detectors.

Developer	Arduino.cc
Type	Single Board Microcontroller
Operating System	None
CPU	Atmel AVR (8-bit), ARM Cortex-M0+ (32-bit), ARM Cortex-M3 (32-bit), Intel Quark (x86) (32-bit)
Memory	SRAM
Storage	Flash,EEPROM

Table 2. Arduino Specifications

Fig 2. Arduino Uno

The Arduino integrated development environment (IDE) is a cross-platform application (for Windows, macOS, Linux) that is written in the programming language Java. It originated from the IDE for the languages Processing and Wiring. It includes a code editor with features such as text cutting and pasting, searching and replacing text, automatic indenting, brace matching, and syntax highlighting, and provides simple one-click mechanisms to compile and upload programs to an Arduino board. It also contains a message area, a text console, a toolbar with buttons for common functions and a hierarchy of operation menus. The source code for the IDE is released under the GNU General Public License, version 2.

The Arduino IDE supports the languages C and C++ using special rules of code structuring. The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output procedures. User-written code only requires two basic functions, for starting the sketch and the main program loop, that are compiled and linked with a program stub main() into an executable cyclic executive program with the GNU toolchain, also included with the IDE distribution. The Arduino IDE employs the program avrdude to convert the executable code into a text file in hexadecimal encoding that is loaded into the Arduino board by a loader program in the board's firmware.

Fig 3. Arduino IDE

3.1.2. SD Card Shield & LCD Display

SD card shield is a data logger which helps to read and write data in .csv file format for further analyses using different tools which use serial communication at 9600 baud rate.It is a versatile device which along with the help of SPI and SD library of Arduino which enables to open and create files.

Fig 4. SD Card Module – Breakout Board

Voltage Level	3.3V,5V
Communication Protocol	SPI Interface
Support	Micro SD, Micro SDHC

Table 3. SD Card Breakout Board Specifications

16X2 LCD Keypad module has following features and is operated by the following algorithms:

- ✓ Blue Backlight with white words
- ✓ uses 4 Bit Arduino LCD Library
- ✓ Left, Right, Up, Down and Select buttons
- ✓ Screen contrast adjustment
- ✓ Arduino Reset button

The board uses different pins to the Arduino example sketches, so to make the display work, use the following sequence of pins when starting the library:
LiquidCrystal lcd(8,9,4,5,6,7);

The buttons are connected to only one analog input pin through resistors to give a different voltage for each button, thus saving on input/output pins.

Fig 5. 16X2 LCD Screen with Keypad

3.1.3. 3D Printing

The 3D printing process builds a three-dimensional object from a computer-aided design (CAD) model, usually by successively adding material layer by layer, which is why it is also called additive manufacturing, unlike conventional machining, casting and forging processes, where material is removed from a stock item (subtractive manufacturing) or poured into a mold and shaped by means of dies, presses and hammers.

The term "3D printing" covers a variety of processes in which material is joined or solidified under computer control to create a three-dimensional object, with material being added together (such as liquid molecules or powder grains being fused together), typically layer by layer. In the 1990s, 3D-printing techniques were considered suitable only for the production of functional or aesthetic prototypes and a more appropriate term for it was rapid prototyping. As of 2019 the precision, repeatability and material range have increased to the point that some 3D-printing processes are considered viable as an industrial-production technology, whereby the term additive manufacturing can be used synonymously with "3D printing". One of the key advantages of 3D printing is the ability to produce very complex shapes or geometries, and a prerequisite for producing any 3D printed part is a digital 3D model or a CAD file.

The most-commonly used 3D-printing process (46% as of 2018) is a material extrusion technique called fused deposition modeling (FDM). While FDM technology was invented after the other two most popular technologies, stereolithography (SLA), and selective laser sintering (SLS); FDM is typically the most inexpensive of the three by a large margin, which lends to the popularity of the process.

The term "3D printing" originally referred to a process that deposits a binder material onto a powder bed with inkjet printer heads layer by layer. More recently, the popular vernacular has started using the term to encompass a wider variety of additive-manufacturing techniques such as electron-beam additive manufacturing and selective laser melting. The United States and global technical standards use the official term *additive manufacturing* for this broader sense.

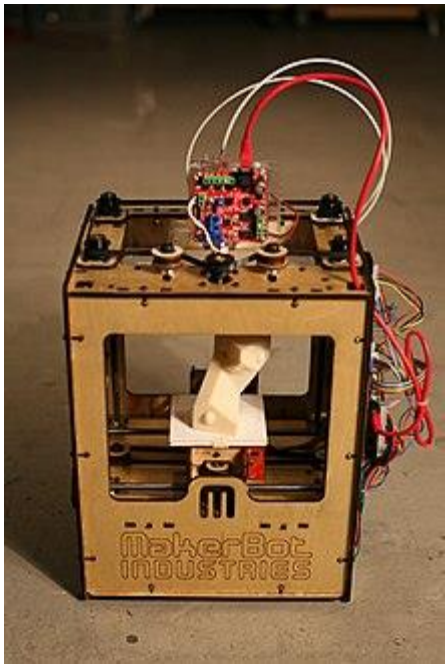

Fig 6. 3D Printer

3.1.4. Solenoid Valve

A **solenoid valve** is an electromechanically operated valve.

Solenoid valves differ in the characteristics of the electric current they use, the strength of the magnetic field they generate, the mechanism they use to regulate the fluid, and the type and characteristics of fluid they control. The mechanism varies from linear action, plunger-type actuators to pivoted-armature actuators and rocker actuators. The valve can use a two-port design to regulate a flow or use a three or more port design to switch flows between ports. Multiple solenoid valves can be placed together on a manifold.

Solenoid valves are the most frequently used control elements in fluidics. Their tasks are to shut off, release, dose, distribute or mix fluids. They are found in many application areas. Solenoids offer fast and safe switching, high reliability, long service life, good medium compatibility of the materials used, low control power and compact design.

Fig 7. 24 V Solenoid Valve

3.2. Design of Rodent Operant Bucket

3.2.1. Introduction

The considerations taken into account while developing the bucket were:

- Robust and ergonomic
- Non-Corrosive
- Electronic Parts should be isolated from fluids
- Fluid Control
- Easy to Operate

3.2.2. Bucket Design & Hardware Installation

There are six key pieces of hardware in ROBucket.The Arduino Uno R3 board is controlled by the ATmega328 microcontroller chip, which has 31.5 kilobytes (KB) of available memory. It has 14 digital I/O pins and six analog input pins. The Arduino Uno board can respond to a single input in less than 1 ms and to multiple conditions in 100–200 ms depending on the task and programming code (D'Ausilio,2012). The SD Card data logging shield works with FAT16 or FAT32 formatted

SD cards, and includes a real-time clock (RTC) to allow accurate date and time stamping of events. The Arduino communicates with the SD card interface of the data logging shield through a serial peripheral interface (SPI), which requires the master output slave input (MOSI); master input slave output (MISO); serial clock (SCK); and chip select (CS) pins (these are digital pins 11, 12, 13, and 10 on the Arduino Uno, respectively). The RTC portion of the data logging shield retrieves and sets time through the I2C bus of the Arduino board. The I2C bus allows multiple devices to communicate with the microcontroller through the same two lines of communication, a clock signal (SCL) and a data line (SDA). The LCD keypad shield has a 16 × 2 character backlight LCD and five keys (Select, Up, Right, Down, and Left).The keypad interface uses analog pin 1 and the LCD screen uses digital pins 4–9 on the Arduino Uno. Digital pin 10 controls the backlight of the screen and can be shared with the data logging shield. The three photo interrupter boards act as the active and inactive nose-poke sensors and a drinking well entry sensor to provide input to the Arduino Uno. The photo interrupter boards have an infrared LED sensor to detect when an object enters the 10 mm gap in the arms at a temporal resolution of 3µs. The Arduino's 5V output signal is detected by the relay module, which controls the opening of a 24V DC solenoid valve. When the solenoid valve is open, sucrose solution is able to flow through the valve into the drinking well inside the chamber. Off the-shelf parts are used to hold the main hardware in place and facilitate their functions.

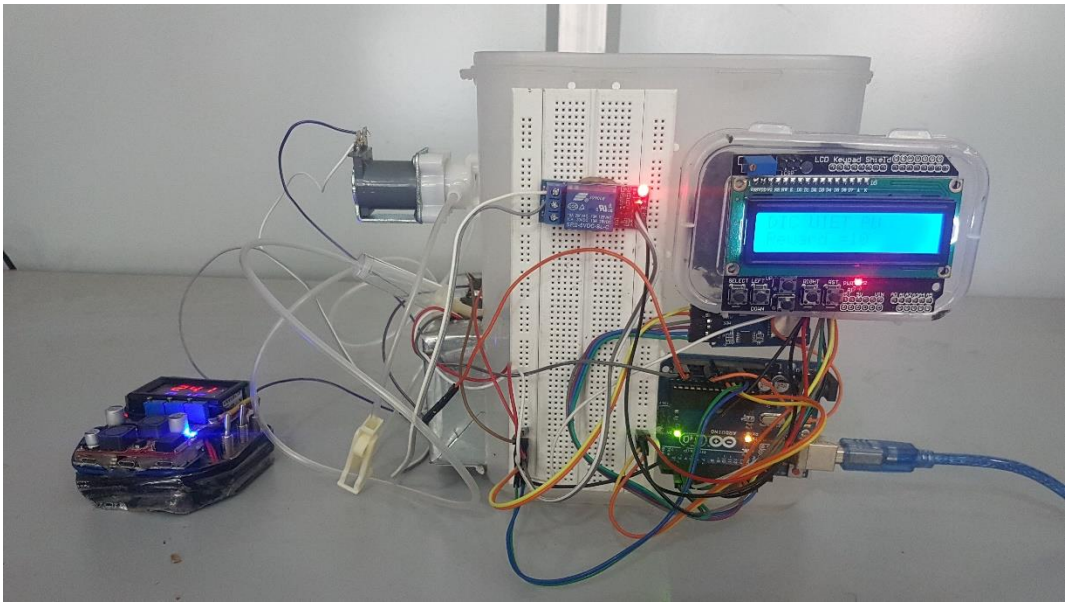

Fig 8. Hardware Setup of RO Bucket

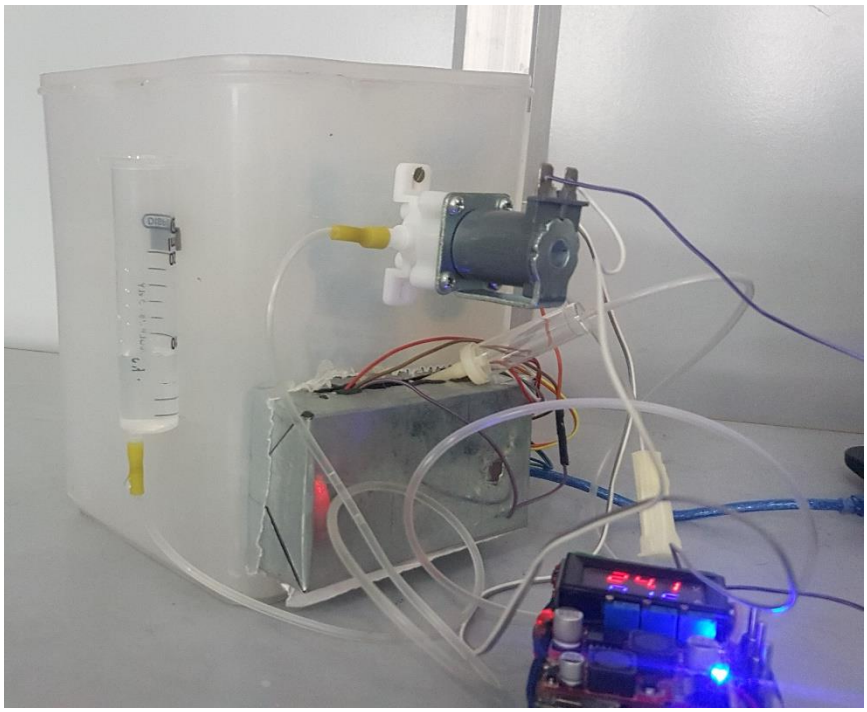

Fig 9. Nose poke Area of RO Bucket

Fig 10. 3D Printed Housing of Nose Poke Area

3.3. Design of Control Mechanism

3.3.1. Introduction

A feedback control system is a system whose output is controlled using its measurement as a feedback signal. This feedback signal is compared with a reference signal to generate an error signal which is filtered by a controller to produce the system's control input.

This devices uses a feedback control system for the operation of solenoid valve using a relay to dispense the liquid. The system is a linear digital two state system which give response signal in form of High or Low.

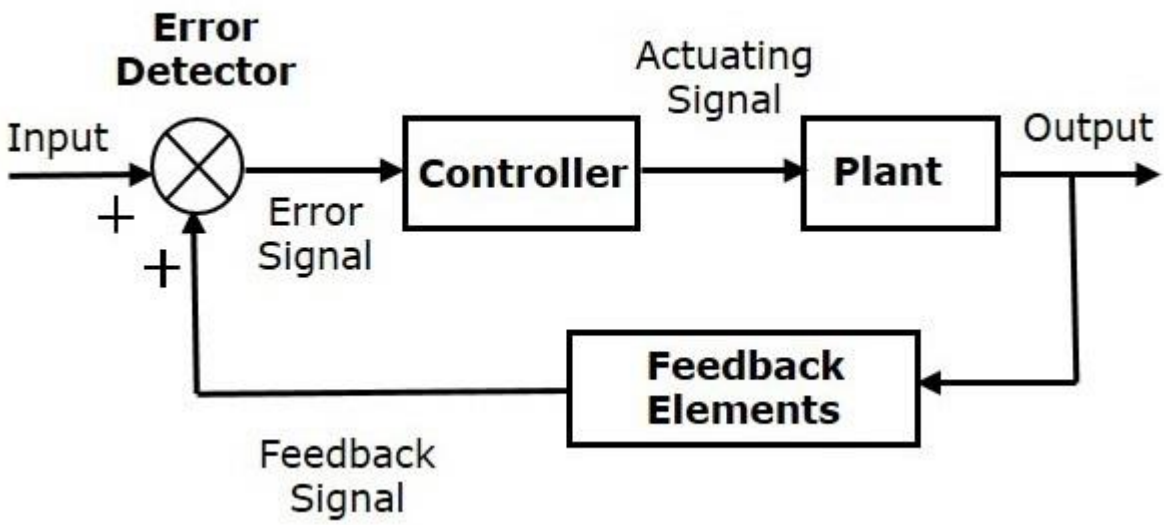

Fig 11. Closed Loop Feedback Control System

3.3.2. Controller Design

Basic operation to be performed by the control system are as follows:

OPERATION	RESPONSE	DATA
Nose Poking in Active	**Delivery of Sucrose**	Recorded
Nose Poking in Inactive	**No Response**	Recorded
Nose Poke in Reward	**No Response**	Recorded

Table 4. Control System Response

Along with these operations the control system also gives command through Arduino to display the required data on the LCD screen as well as record the data.

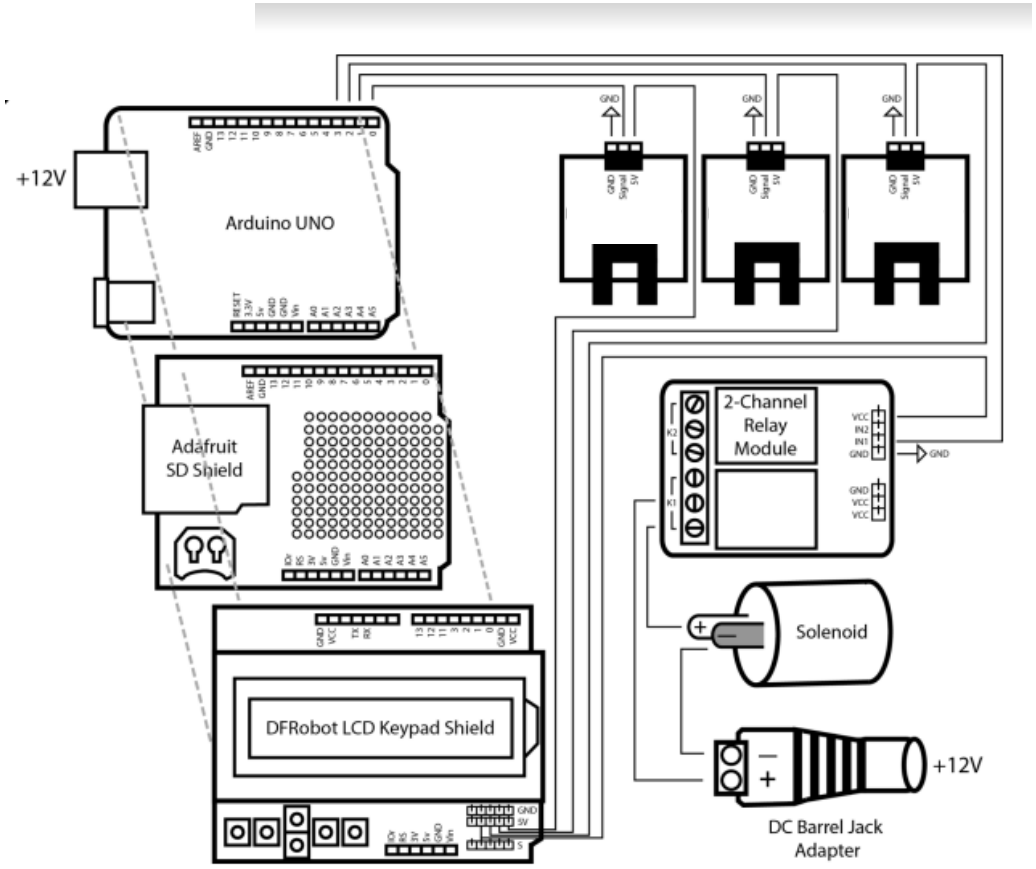

Fig 12. Circuit Diagram of RO Bucket

3.4. Design of Data Logger

3.4.1. Introduction

A data logger (also datalogger or data recorder) is an electronic device that records data over time or in relation to location either with a built in instrument or sensor or via external instruments and sensors. Increasingly, but not entirely, they are based on a digital processor (or computer). They generally are small, battery powered, portable, and equipped with a microprocessor, internal memory for data storage, and sensors. Some data loggers interface with a personal computer, and use software to activate the data logger and view and analyze the collected data, while others have a local interface device (keypad, LCD) and can be used as a stand-alone device.

Data loggers vary between general purpose types for a range of measurement applications to very specific devices for measuring in one environment or application type only. It is common for general purpose types to be programmable; however, many remain as static machines with only a limited number or no changeable parameters. Electronic data loggers have replaced chart recorders in many applications.

One of the primary benefits of using data loggers is the ability to automatically collect data on a 24-hour basis. Upon activation, data loggers are typically deployed and left unattended to measure and record information for the duration of the monitoring period. This allows for a comprehensive, accurate picture of the environmental conditions being monitored, such as air temperature and relative humidity.

The cost of data loggers has been declining over the years as technology improves and costs are reduced. Simple single channel data loggers cost as little as $25. More complicated loggers may costs hundreds or thousands of dollars.

3.4.2. Data Logger

Data logger in the RO Bucket records and displays data in different mediums:

- Records data in .csv file in SD Card
- Displays Data on 16X2 LCD Screen
- Displays Real-time Data on serial Monitor of the computer
- After the process completion displays data in a tabular format

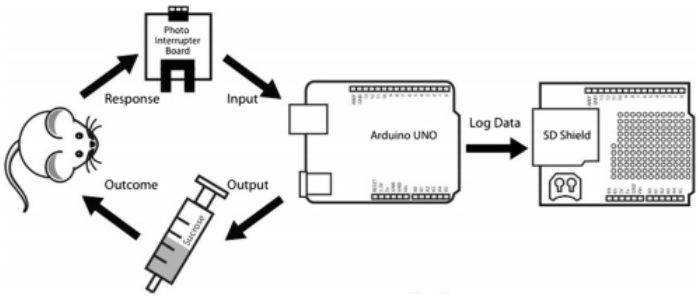

Fig 13. Data Logger Block Diagram

Chapter 4. Performance of System in Real Time Environment

4.1. Introduction

Performance Analysis is a specialist discipline involving systematic observations to enhance performance and improve decision making, primarily delivered through the provision of objective statistical (Data Analysis) and visual feedback (Video Analysis).

The performance of the device in real-time environment is very essential for the developer to check out for problems and get feedback for its refinement of results. The system records data and conveys the output using LCD screen, Serial Monitor and ThingSpeak.

The performance of system is measured in an artificial environment without any live mice and the data is analyzed. The system gave optimum and required results most of the time and has a very reliable and robust architecture.

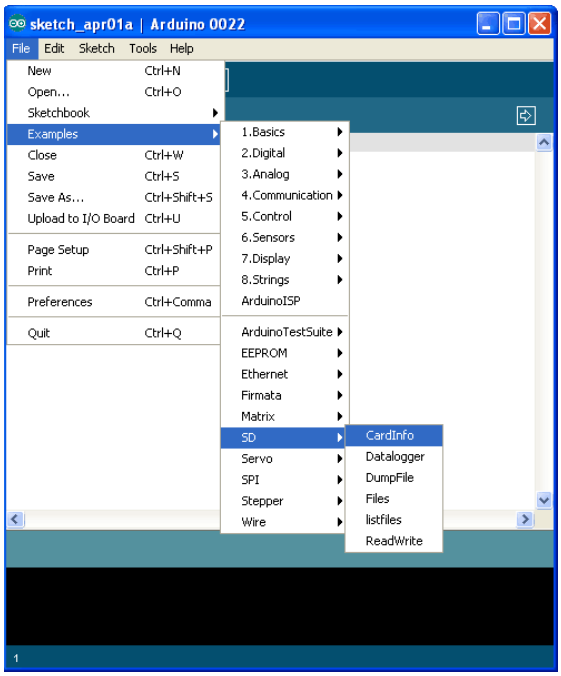

Fig 14. SD Card Data Logger in Arduino

4.2. Results and Discussions

The following points are to be noted for performance analysis of the RO Bucket:

- The IR sensors gave a satisfactory response of the movement of the rodent and was able to detect even the slightest of the activity in the pokes of the device.
- Solenoid was able to dispense the correct amount i.e. 60µL of fluidic solution upon triggering by a relay on a delay on 10 ms.
- Data logging system of the device was able to record ample amount of data for a detailed and proper analysis of the operational behavior of the mice at a baud rate of 9600.
- Algorithm of the control mechanism was efficient and well within time constraints required for a good feedback control system with a time complexity of 1.2.

The results of the test case performed by our team are given as follows:

```
Time,Active,Reward,Passive
26/11/2019,10:20,20,9,5
26/11/2019,10:37,30,15,10
26/11/2019,10:37,12,4,10
```

Fig 15. CSV File in SD Card

Active	Rewards	Passive	Cases
20	9	5	1(Normal)
30	15	10	2(Addicted)
12	4	10	3(Alzheimer)

Fig 16. Sample Data

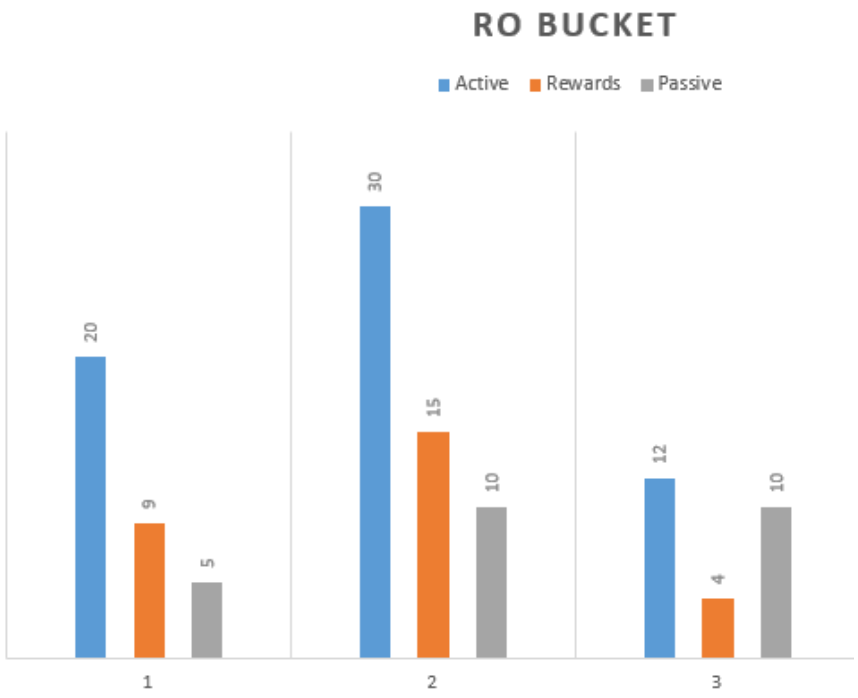

Fig 17. Graphical Representation of Sample Data in ThingSpeak

Chapter 5. Conclusions and Future Scope

5.1. Conclusions

After the performance analysis of the data and the sketching of results is done. By the trends shown in the results, it is observed that all the objectives of the system are achieved.The overall cost of the device is reduced under INR 2000.

Hence, A reliable and robust low cost RO Bucket is conceptuliazed.

5.2. Future Scope

- A better sensor than IR sensor would be installed to better analyze the data.
- AI and ML algorithms would be used to detect the behavior with the help of camera.
- Live Monitoring of data through internet would also be facilitated and live graphs would also be displayed.
- Aesthetics of the whole equipment would be well planned and design would be 3D printed using a printer.
- PCB of the whole circuit would be conceptualized.
- Further cost reduction parameters would be incorporated.

Appendix

1. Arduino Code

```
#include <LiquidCrystal.h>
#include <SPI.h>
#include <SD.h>

const int chipSelect = 3;
const int ac_poke = A5;
const int rc_poke = A4;
const int pc_poke = A3;
const int relay=2;
const int pin_RS = 8;
const int pin_EN = 9;
const int pin_d4 = 4;
const int pin_d5 = 5;
const int pin_d6 = 6;
const int pin_d7 = 7;
const int pin_BL = 10;
int sv1 =0 ;
int sv2 =0 ;
int sv3 =0 ;
int ac=0;
int rc=0;
int pc=0;
```

```
int ac_state=0;
int rc_state=0;
int pc_state=0;
int x;
LiquidCrystal lcd( pin_RS,  pin_EN,  pin_d4,  pin_d5,  pin_d6,  pin_d7);
void setup() {
 pinMode(2, OUTPUT);
 lcd.begin(16, 2);
 lcd.setCursor(0,0);
 lcd.print("DIC UIET PU");
 lcd.setCursor(0,1);
 lcd.print("READY");
 Serial.begin(9600);
  /*while (!Serial) {
    ; // wait for serial port to connect. Needed for native USB port only
  }


  Serial.print("Initializing SD card...");

  // see if the card is present and can be initialized:
  if (!SD.begin(chipSelect)) {
    Serial.println("Card failed, or not present");
    // don't do anything more:
    while (1);
  }
  Serial.println("card initialized.");*/
```

```
}
void loop() {
 /* String dataString = "";

  // read three sensors and append to the string:
  for (int analogPin = 0; analogPin < 3; analogPin++) {
    int sensor = analogRead(analogPin);
    dataString += String(sensor);
    if (analogPin < 2) {
      dataString += ",";
    }
  }

  // open the file. note that only one file can be open at a time,
  // so you have to close this one before opening another.
  File dataFile = SD.open("datalog.txt", FILE_WRITE);

  // if the file is available, write to it:
  if (dataFile) {
    dataFile.println(dataString);
    dataFile.close();
    // print to the serial port too:
    Serial.println(dataString);
  }
  // if the file isn't open, pop up an error:
  else {
    Serial.println("error opening datalog.txt");
```

```
}*/
sv1 = analogRead(ac_poke);
sv2 = analogRead(rc_poke);
sv3 = analogRead(pc_poke);
Serial.print("\nactive = ");
Serial.print(sv1);
Serial.print("\nreward = ");
Serial.print(sv2);
Serial.print("\npassive = ");
Serial.print(sv3);
if(sv1 < 800){ //checks if object is there or not
  Serial.print("\nObject Detected");
  ac+=1;
  delay(1000);
  }
if(sv2 < 800){ //checks if object is there or not
  Serial.print("\nObject Detected");
  rc+=1;
  delay(1000);
  digitalWrite(2, HIGH);
  delay(10);
  digitalWrite(2, LOW);

  }
if(sv3 < 100){ //checks if object is there or not
  Serial.print("\nObject Detected");
  pc+=1;
```

```
  delay(1000);
  }
 else{
  Serial.print("\nNo object in Front");
  }
x = analogRead (0);
lcd.setCursor(0,1);
if (x < 60) {
 lcd.print ("Operating");
}
else if (x < 200) {
 lcd.print ("Reward =");
 lcd.print (rc);
}
else if (x < 400){
 lcd.print ("Passive =");
 lcd.print (pc);
}
else if (x < 600){
 lcd.print ("Active =");
 lcd.print (ac);

}
else if (x < 800){
 ac =0;
 pc=0;
 rc=0;
```

```
    }
  }
```

2. Google Charts Using ThingSpeak

Google Doc is increasing and we forecast a tremendous growth in the use of cloud office suites in the near future. So we are starting a separate category on this site particularly for Google Docs and kicking off with our first Google Doc tutorial that about creating a line chart or line graph in Google Doc Spreadsheet.

Statistical graphs have many advantages. On Google Spreadsheet you can create different types of graphs including Line, Area, Column, Bar, Pie, Map etc. Each graph types used for different purposes. A Line chart can be used to compare multiple continues data sets that means it's suitable for showing change over time.

The procedure to create a line chart or line graph in Google Doc Spreadsheet is similar to the procedure we follow in Microsoft Office Excel Charts. Because of this similarity, those who're familiar with Microsoft Excel Spreadsheet may find it pretty easier in creating charts in Google Doc Spreadsheet.

Steps for using Google Charts

1. Select the entire data, then go to the "Insert" menu and select "Chart". It will open the Chart Editor dialogue box.
2. Click on the tab labeled as "Charts". Then click "Line". There you can see different Line charts. Choose the first one that is more common.
3. Chart Editor Dialogue Box in Google Spreadsheet
4. Now the chart customization part is only left. For this click the tab "Customize".
5. Give a title for your Chart. Then choose legend position.
6. In the above both the cases, you can change the font color and font size, if you wish. Now choose a "Feature".
7. Titling and Labeling on charts in Google Spreadsheet

8. Now Choose See Axis customization options there.
9. Label horizontal and vertical axis. The horizontal axis shows the overs in the match so label it as “Overs”. Y-axis label as “Runs”.

Chart Editor Customization tips

You are almost finished creating your first chart in Google Doc Spreadsheet. Now if you wish you can change the background color, grid lines etc. from the same dialogue box and click “Update”.

ThingSpeak Key Features

- ThingSpeak allows you to aggregate, visualize and analyze live data streams in the cloud. Some of the key capabilities of ThingSpeak include the ability to:
- Easily configure devices to send data to ThingSpeak using popular IoT protocols.
- Visualize your sensor data in real-time.
- Aggregate data on-demand from third-party sources.
- Use the power of MATLAB to make sense of your IoT data.
- Run your IoT analytics automatically based on schedules or events.
- Prototype and build IoT systems without setting up servers or developing web software.
- Automatically act on your data and communicate

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